

YIZHENG LU

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EDUCATION

University of Oxford Sep 2025 - Aug 2026

M.Sc. in Mathematical and Computational Finance (GPA: 74/100, Distinction, **Top 10%**)

Key courses: Stochastic Calculus, Financial Derivatives, Numerical Methods (Monte Carlo & FDM), C++, Deep Learning, DeFi, Market Microstructure, Fixed Income, Risk Management

University of Liverpool (2+2 program with XJTLU) Sep 2021 - Aug 2025

B.Sc. in Mathematics (GPA: 91/100, First Class Honours, **Ranked: 1/275**)

Key courses: Financial Mathematics, Measure Theory & Probability, Stochastic Process, Machine Learning and Data Science, ODE & PDE

RESEARCH INTEREST

Mathematical Finance, Financial Engineering, FinTech

Probability Theory, Probabilistic Machine Learning, Deep Learning

SELECTED RESEARCH EXPERIENCE

Yizheng Lu, supervised by Prof. Blanka Horvath and Dr. Katia Babbar (University of Oxford).¹
Institutional Trading Constraints and the Implied Volatility Surfaces of IBIT and Bitcoin Options.

- Developed an arbitrage-free **SVI-JW and SANOS (Smooth Arbitrage-free Non-parametric Surfaces)** calibration pipeline for **IBIT and Deribit Bitcoin options**, accounting for the early-exercise feature of American-style IBIT puts to compare implied-volatility surfaces.
- Analysed minute-level Bitcoin returns across U.S. trading, overnight, and weekend periods; two-sample Kolmogorov–Smirnov tests found significant differences in hourly return distributions.
- Derived a discrete-hedging framework and ran 74 weekly delta-hedging simulations. Restricting rebalancing to NASDAQ hours increased per-cycle P&L standard deviation by 34.4% relative to continuous hedging, while the difference in mean P&L was statistically insignificant.
- Found an average IBIT at-the-money implied-volatility premium of approximately 2.8–3.2 percentage points across the tenors studied. Tested a cross-market convexity trade and found that estimated transaction costs eliminated its gross advantage.

Yizheng Lu, supervised by Prof. Justin Sirignano (University of Oxford). *Neural Network Surrogates for Option Pricing and Heston Model Calibration.*

- Developed a **PyTorch-based neural network surrogate** for European call pricing under the **Heston stochastic volatility model**, using 50,000 Monte Carlo price samples; achieved a mean relative error of 2.25% on 10,000 test set samples.
- Compared nine network depth–width configurations and analysed pricing errors across moneyness and maturity, identifying larger relative errors for out-of-the-money and shorter-maturity options.
- Implemented calibration of five model parameters through automatic differentiation and Adam optimisation over the trained pricing surrogate, fitting a synthetic surface of 800 contracts and achieving a pricing MAE of 0.1714 on 100 additional contracts.

¹MSc dissertation at Oxford, undertaken during a Quantitative Researcher internship at Immersive Finance (London).

- Examined discrepancies between price fit and parameter recovery, finding that low pricing errors could coexist with substantial parameter estimation errors and highlighting limitations of surrogate-based calibration.

Yizheng Lu, supervised by Dr. Youness Boutaib (University of Liverpool). *Neural Networks for Financial Time-Series Classification*.

- Derived backpropagation and backpropagation-through-time mathematical formulation using matrix calculus, and examined vanishing and exploding gradients in recurrent networks.
- Implemented and compared feed-forward, RNN, LSTM and GRU architectures across MNIST and Japanese Vowels benchmarks; achieved approximately 95% test accuracy with LSTM on speaker classification.
- Formulated S&P 500 return prediction as a nine-class classification task using 13 market, macroeconomic, and technical features; compared equal-width and equal-frequency discretisation and evaluated RNN and LSTM models to investigate class imbalance and differences in market predictive performance.

PROFESSIONAL EXPERIENCE

Guotai Haitong Securities
Quantitative Researcher Intern

Jun 2025 - Aug 2025
Suzhou, CN

- Investigated a dynamic leverage strategy using drawdown and recovery signals within a risk-parity portfolio of 20 U.S. ETFs; developed a rolling backtesting framework to evaluate performance against the S&P 500.
- Implemented numerical pricing algorithms for equity and interest-rate derivatives in C++ using the firm's in-house library.

Bank of Suzhou
Financial Data Analyst Intern

Jan 2023 - Mar 2023
Suzhou, CN

- Processed customer datasets and engineered predictive features for classification models in Python.
- Compared logistic regression, decision trees, and support vector machines, with the best-performing model achieving 89% accuracy and an F1 score of 0.86.

AWARDS

JP Morgan 2025 Quantitative Trading Challenge (London) Excellence Award	Oct, 2025
Willis Prize in Mathematics (Top 1%), University of Liverpool	Jul, 2024
Exchange Student Excellence Award (Top 1%), University of Liverpool	Jul, 2023
Second Prize Winner, Asia and Pacific Mathematical Contest in Modelling (APMCM)	Nov, 2022
Academic Excellence Award (Top 1%), Xi'an Jiaotong Liverpool University	Aug, 2022
Honorable Mention, Mathematical Contest in Modelling (MCM & ICM)	Feb, 2022

SKILLS

Agent Usage: Codex, Claude Code

Programming: Python (PyTorch, Numpy, Pandas, Scikit-learn), C++, R, SQL

INTERESTS

Football, basketball, badminton, pool and snooker, Texas Hold'em poker, travel, and photography.